Retail Sales Performance Analysis

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POWER BI | 2025 SALES PERFORMANCE | PORTFOLIO CASE STUDY

An interactive, two-page report designed to communicate retail sales performance, profitability and product and regional trends.

Headline result: $541.93K revenue and a 39.81% profit margin (simplified measure) in 2025. East is the top region, Printer Paper the top product and June the peak sales month.

Project at a glance

Project type	Retail sales analytics and dashboard development
My role	Sole analyst: data preparation, modelling, DAX measures and dashboard design
Tool	Microsoft Power BI; Power Query; DAX
Reporting period	2025
Deliverables	Executive Overview; Product & Regional Analysis
Data	Practice retail dataset; synthetic transactions

Business problem

The retail business needs a clear view of sales performance across products, regions and sales channels. Without consolidated performance reporting, managers may struggle to monitor revenue, assess profitability, identify trends and evaluate underperforming areas. This project transforms transactional data into an interactive report to support informed decisions.

Project objectives

- Present key performance indicators in a clear, interactive executive view.
- Compare revenue across regions, products and sales channels.
- Evaluate profitability across product categories.
- Identify monthly sales trends and areas requiring further investigation.

Data and methodology

Data preparation

Used Power Query to standardize inconsistent text values and improve reporting consistency, including category and sales-channel labels. A small Unknown sales-channel share remains visible in the report and is acknowledged as a data-quality limitation.

Data modelling

Built relationships between SalesTransactions and lookup tables for Products, Customers and Regions, using common IDs. A dedicated DateTable supports analysis by year and month and is related to transaction OrderDate.

DAX measures

Core measures include Total Revenue (SUMX of Quantity × UnitPrice), Total Profit (SUMX of (UnitPrice − UnitCost) × Quantity), Total Orders (DISTINCTCOUNT), Units Sold (SUM), and Profit Margin (DIVIDE of Total Profit by Total Revenue). The profit measure is a simplified product-cost-based measure; it does not deduct discounts or operating expenses.

Dashboard 1 | Executive Overview

A high-level management view of revenue, profit, margin, orders, units sold, monthly revenue trends and sales-channel mix. Region, sales channel and year controls support interactive filtering.

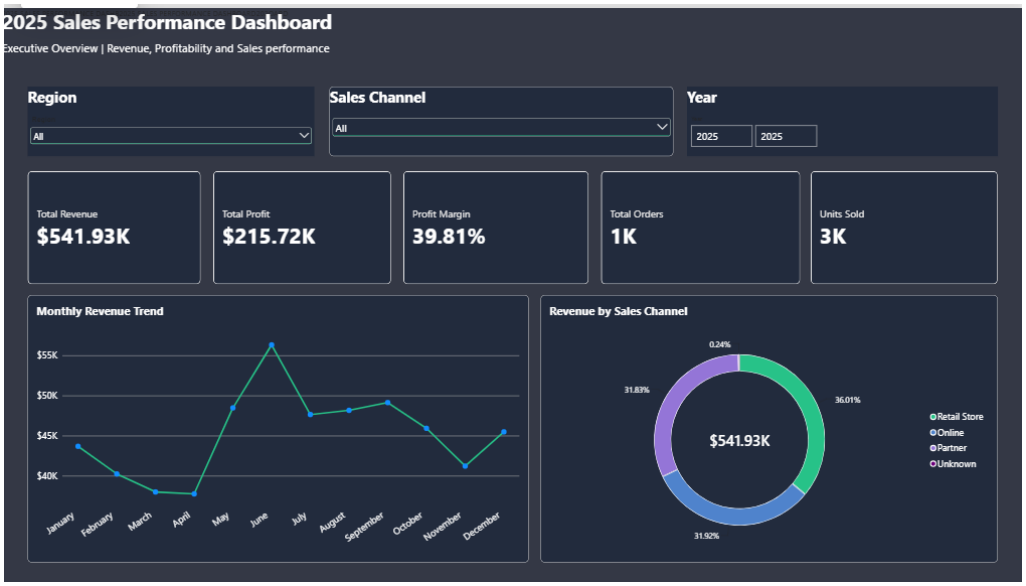


Figure 1. Executive Overview dashboard (all regions and channels selected).

Key performance indicators

Headline figures: total revenue $541.93K; total profit $215.72K (simplified measure); profit margin 39.81%; ~1K orders and ~3K units sold (rounded on the dashboard cards).

June shows the highest revenue on the monthly trend. Retail Store represents the largest sales-channel share at 36.01%, followed by Online at 31.92% and Partner at 31.83%. Unknown accounts for approximately 0.24%.

Dashboard 2 | Product & Regional Analysis

A more detailed view designed to identify leading regions, top-selling products and category-level profitability.

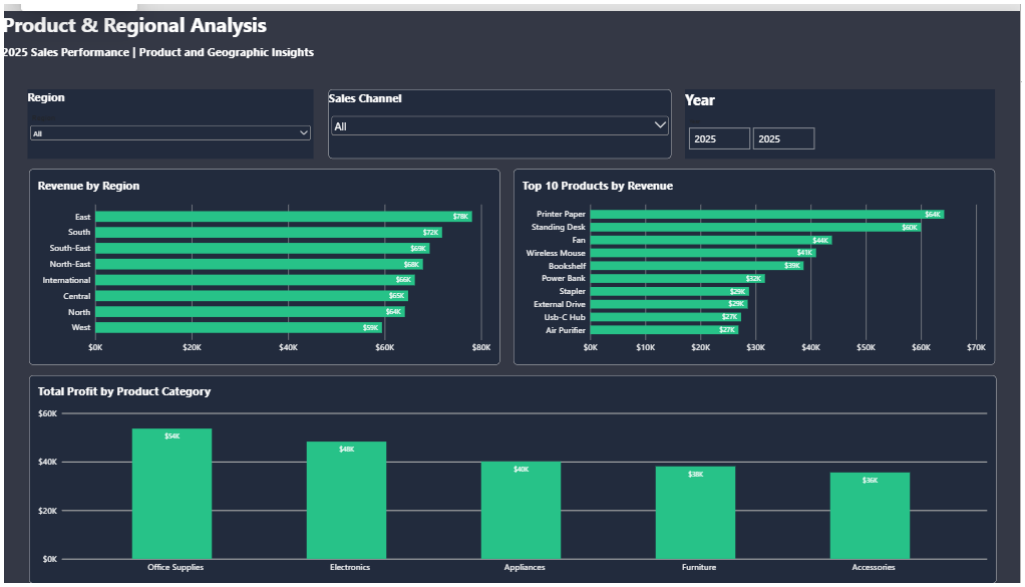


Figure 2. Product & Regional Analysis dashboard (all regions and channels selected).

Findings from the dashboard

- East leads regional revenue at approximately $78K.
- Printer Paper leads product revenue at approximately $64K.
- Office Supplies records the highest category profit under the project measure, at approximately $54K.
- Regional and product performance varies, creating opportunities for targeted investigation.

Business interpretation and recommendations

The East region's leading revenue position is a finding, not proof that it should receive additional investment immediately. Investigate its product mix, demand, profitability and operating context, then assess whether successful approaches could transfer to other regions.

Office Supplies contributes the largest measured category profit, but concentrating investment in one category would create risk. Evaluate demand, margins, capacity and broader operating costs before reallocating resources.

June's revenue peak should be explored using order counts, units sold, average order value and product mix. The monthly revenue chart alone does not establish the cause of the increase.

Challenges and solutions

Dashboard layout and formatting

Arranging multiple KPI cards, slicers and charts in a limited report space was the main challenge. The layout was refined through resizing, alignment, spacing, colour adjustments and separation into two pages with distinct analytical purposes.

DAX learning curve

Developing confidence with DAX was another challenge, especially understanding when to use row-wise expressions and how filter context affects measures. Practical exercises and iterative testing improved understanding of SUMX, CALCULATE and DIVIDE.

Limitations and analytical care

- The dataset is synthetic and intended for skills demonstration rather than real company reporting.
- Figures are transcribed from dashboard visuals; rounded chart labels are approximate.
- The Total Profit calculation subtracts recorded unit cost but does not incorporate discounts or operating expenses, so it should not be described as net profit.
- The Unknown sales-channel segment indicates a remaining classification issue.
- The dashboard shows patterns and comparisons, not verified causal explanations.

Conclusion

This project demonstrates how an interactive Power BI dashboard can turn retail transaction data into understandable performance information. It highlights monthly revenue trends, regional and product differences, and category profitability while showing why business recommendations require careful investigation. The work strengthened practical skills in data cleaning, modelling, DAX, dashboard formatting and analytical reasoning.

Skills demonstrated

Power Query • Data modelling • DAX measures • KPI design • Data visualization • Business interpretation • Dashboard usability